

Riya Fungal Biology Creative Assignment

This project is a three-dimensional model representing the parasitic fungus *Cordyceps* infecting an ant host. The model depicts a forest floor environment, with an infected ant positioned on the substrate and a fungal fruiting body emerging from its body.

The aim of this model is to educate viewers about parasitic fungi and their ability to manipulate host behaviour, a concept that is often unfamiliar to the general public. This project seeks to address “fungal blindness” by showcasing a dramatic and memorable example of fungal ecology.

Through this model, viewers can better understand how fungi are not only decomposers but also active biological agents that influence ecosystems through parasitism and host interactions.

Cordyceps is a genus of entomopathogenic fungi, meaning it infects and parasitises insects and other arthropods. The infection begins when fungal spores attach to and penetrate the host’s exoskeleton. Once inside, the fungus proliferates as hyphae, consuming the host’s internal tissues.

A key feature of *Cordyceps* infection is its ability to alter host behaviour. In infected ants, the fungus induces a phenomenon often referred to as “summit disease,” where the ant is compelled to climb vegetation and attach itself in an elevated position. This behaviour enhances the dispersal of fungal spores.

After the host dies, the fungus continues to grow and produces a fruiting body (stroma) that emerges from the host’s body. This structure releases spores into the environment, allowing the fungal life cycle to continue.

This interaction demonstrates parasitism, where the fungus benefits at the expense of the host, and highlights the ecological role of fungi in regulating insect populations and contributing to biodiversity.

The *Cordyceps*-host interaction illustrates the complexity of fungal life cycles and their ecological significance. Such fungi play a role in controlling insect populations and maintaining ecological balance.

Additionally, studying *Cordyceps* has broader scientific implications. Research into host manipulation provides insights into neurobiology, parasitology, and evolutionary biology. Certain species of *Cordyceps* are also of interest in medicine and biotechnology due to their bioactive compounds.

This model highlights the fascinating and often overlooked role of fungi as dynamic organisms capable of complex interactions with other living beings. By presenting *Cordyceps* in a visual and engaging format, this project aims to increase awareness and appreciation of fungal biology and its importance in natural ecosystems.



References

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